DRIVE Trial

Design Selection

Introduction

- Reducing amount of time spent driving from home to work can help maximize productivity in the workplace.

- Compare designs to determine the optimal driving route from participants home to his workplace.

- Three route option: Plaza (control), Rainbow, Stateline.

Overview

Primary Endpoint:

- Driving time (in minutes)

Secondary Endpoints:

- No. of times the route changed
- No. of serious adverse events
- No. of minor adverse events.
- Weather condition

Design Groups

- Fixed Design
- RAR, lower interims
- RAR, arm dropping
- RAR, higher interims
- RAR, no fixed allocation to Control

**All designs considered Independent dose model (See SAP for details).*

Fixed Design

- Deterministic accrual, no dropouts.
- 1:1:1 fixed allocation.
- No interims, final analysis at 33.
- A trial was declared successful if posterior probability of Stateline or Rainbow drive time being less than the Control drive time by 1 minute was greater than 0.7.
- The route to compare to the Control was picked based on the maximum posterior probability of Rainbow or Stateline having the lowest time.
- The prior mean μ_d was set to 15 and prior standard deviation τ_d^2 was set to 6. For variance prior σ_μ was set to 1.2 and σ_n was set to 0.1.

RAR, lower interims

- Adaptive design with response adaptive randomization (RAR).
- Deterministic accrual, No dropout rate.
- Final endpoint = 0.01 weeks.
- Externally predefined accrual file.
- Burn-in allocation ratio: 2:1:1 then RAR. 25% allocation to the control, Block size = *4*
- Interims *3* at *8,16,24* and final analysis at 33.
- The study would stop for futility if posterior probability of Stateline or Rainbow drive time being less than the Control drive time by 1 minute was less than 0.1.
- The study would stop for success if posterior probability of Stateline or Rainbow drive time being less than Control drive time was greater than 0.998.
- The final success of a trial was determined by if posterior probability of Stateline or Rainbow drive time being less than Control drive time was greater than 0.9836.
- The prior mean μ_d was set to 15 and prior standard deviation was set τ_d^2 to 5. For variance prior σ_μ was set to 5 and σ_n was set to 0.1.

RAR, arm dropping

- Arm dropping design.
- One interim at 15 and final analysis at 33.
- Only futility criteria.
- Optimistic prior.
- If the posterior probability of having the smallest driving time is less than 0.5, Rainbow or Stateline arm would be dropped.
- The study would be declared futile if at final evaluation if the posterior probability of Stateline or Rainbow drive time being less than Control drive time was less than 0.4.
- The study would be declared successful if the posterior probability of Stateline or Rainbow drive time being less than Control drive time was greater than 0.98.
- The prior mean μ_d was set to 15 for Control and 13 for Rainbow and Plaza routes; and prior standard deviation was set τ_d^2 to 3 in all arms. For variance prior σ_μ was set to 10 and σ_n was set to 0.1.

RAR, higher interims

- Deterministic accrual.
- Burn-in allocation ratio 4:3:3 then RAR. Block size = 10, 40% allocated to control.
- Interims 5 at 10,15,20,25,30 and final analysis at 33.
- Optimistic prior
- The study would stop for futility if posterior probability of Stateline or Rainbow drive time being less than Control drive time by 0.5 minute was less than 0.15.
- The study would stop for success if posterior probability of Stateline or Rainbow drive time being less than Control drive time was greater than 0.99925.
- The final success of a trial was determined by the posterior probability of Stateline or Rainbow drive time being less than Control drive time was greater than 0.99.
- The prior mean μ_d was set to 10 and prior standard deviation was set τ_d^2 to 10. For variance prior σ_μ was set to 1.2 and σ_n was set to 0.01.

RAR, no fixed allocation to Control

- Deterministic accrual.
- Final endpoint 0.01 weeks.
- Burn-in allocation ratio 4:3:3 then RAR. Block size = 10
- Under no fixed allocation to the Control all three arms would be considered for response adaptive randomization.
- Interims 5 at 10,15,20,25,30 and final analysis at 33.
- The study would stop for futility if posterior probability of Stateline or Rainbow drive time being less than Control drive time by 0.5 minute was less than 0.15.
- The study would stop for success if posterior probability of Stateline or Rainbow drive time being less than Control drive time was greater than 0.99925.
- The final success of a trial was determined by the posterior probability of Stateline or Rainbow drive time being less than Control drive time was greater than 0.99.
- The prior mean μ_d was set to 10 and prior standard deviation was set τ_d^2 to 10. For variance prior σ_μ was set to 1.2 and σ_n was set to 0.01.

Initial Designs & Results

Design Type	Number of Interims (at N drives)[1]	Early Stopping Criteria	Final Criteria	Scenario (effect sizes)	Mean N Drives	Type 1 Error Rate/ Power
Fixed	0	None	Success	Null	33.0	0.025
				Expected	33.0	0.978
				Best	33.0	0.934
RAR, lower interims	3 (**8**, 16, 24)	Futility and Success	Success	Null	23.6	0.024
				Expected	23.6	0.980
				Best	25.1	0.944
RAR, arm dropping	1 (**15**)	Futility	Futility and Success	Null	30.5	0.023
				Expected	32.9	0.987
				Best	32.9	0.932
RAR, higher interims	5 (**10**,15,20,25,30)	Futility and Success	Success	Null	25.6	0.020
				Expected	22.5	0.978
				Best	24.1	0.935
RAR, no fixed allocation to Control	5 (**10**,15,20,25,30)	Futility and Success	Success	Null	25.7	0.026
				Expected	24.3	0.902
				Best	25.4	0.886

[1]All bolded N values represent when we start to evaluate the early stopping rules.

Final Design: RAR, lower interims

- Best operating statistics.
- Less interim analysis.
- The power of the one best scenario is 94.4%, which is superior to any other design's power.
- The expected number of drives under all scenarios are the smallest (23.6, 23.6, and 25.1), which reduces the trial duration.